Lars Larsen

Hardware Test, Verification & Engineering Tools Engineer

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SUMMARY

Engineer who builds the systems that make hardware teams fast and correct: hardware-in-the-loop video test rigs, bit-exact instrument models, scope automation, simulation reviews, an in-house KiCad fork and the PLM/ERP integrations behind them. Lead designer of LZX Industries' 100+ products since 2011.

EXPERIENCE

Founder & Principal Engineer, Verification & Tooling — LZX Industries

Portland, OR · 2011 – Present

Video synthesizer and video-processing instrument maker (DBA 2011, LLC 2017). Lead designer of every product in a three-person core team (partners on firmware/HDL and board layout).

- Led 100+ hardware products from concept to production across six Eurorack series, standalone instruments, DIY kits, cases and power: ~240 production circuit-board designs and ~140 metal front panels; $6.4M direct-to-consumer sales (2018–2026) plus a dealer channel.
- Videomancer verification (2026): one analysis library judges GHDL simulation and real HDMI captures (SSIM/PSNR); 984-case sweep tier; 46 named regression guards for release-candidate defects.
- Chromagnon (2026): Python instrument model bit-exact to the iCE40 RTL and GHDL, with nine block test runners (76/76 injected faults caught in simulation), Rigol MSO5000 automation over SCPI, and 88 golden-vector tests for its WebGL browser port.
- Simulation (2026, AI-assisted tooling): ngspice reviews with derating rules and IBIS-derived driver models — a ~200-check pre-fab review of a clock board found 9 failures fixed before fabrication; ~60-part shared model library.
- Engineering tools (2026, AI-assisted): C++ KiCad fork (LZX Designer) with .lzx formats, parametric Eurorack templates and an InvenTree parts library; Python pipelines that generate schematics, placement, routing and silkscreen; a parametric sheet-metal case generator.
- Software: led the Shopify Hydrogen/React storefront since 2023 (~58k lines, Playwright); in 2026, with AI coding agents, a Python operations CLI (~1,000 commands, ~3,500 tests) integrating Shopify, InvenTree, ERPNext, JLCPCB and UPS, a Rust/Tauri firmware updater, and self-hosted Forgejo CI and Authentik SSO.

Operations Manager — Darkplace Manufacturing

Portland, OR · 2015 – 2017

Manufacturing arm for the LZX Industries product line.

- Managed operations, purchasing and customer communication for a manufacturing team of 10+ people building the LZX product line, while continuing as its product designer.

Earlier

- Multimedia Programmer, University of North Texas, Denton TX — 2008–2010, 2011–2015
- Lead Service Technician, Switched On (synthesizer sales & repair), Austin TX — 2010
- New Media Developer, FUNimation Entertainment, Flower Mound TX — 2007–2008
- Multimedia Developer, TargetScope, Plano TX — 2006–2007

SELECTED PROJECTS

- **Videomancer SDK (GPL-3.0)** — iCE40 video platform, program ABI and signed package format, GHDL image tester; ~190 C++ and ~350 VHDL unit tests. github.com/lzxindustries/videomancer-sdk
- **Chromagnon web simulator** — In-browser WebGL port of the Chromagnon instrument model, checked against golden vectors. lzxindustries.net/instruments/chromagnon/tools/simulator
- **LZX Industries storefront** — Shopify Hydrogen/React storefront with instrument hubs, firmware downloads and live tools. lzxindustries.net

SKILLS

Software	Python (NumPy, OpenCV, PySide6), TypeScript/React (Shopify Hydrogen/Remix, Vitest, Playwright), Docker, Forgejo CI, Linux servers, AI-assisted development (Claude Code, Cursor, MCP)
Firmware & FPGA	C/C++ (C++17/20, freestanding, no-heap), Zynq-7000 bare-metal and FreeRTOS, STM32F4, RP2040/RP2350, AVR, TinyUSB, Ed25519 signing; VHDL, Verilog, GHDL, VUnit, Yosys/nextpnr (iCE40; ECP5 designs), Vivado (Zynq-7000)
Electronics	Analog/mixed-signal and video (SD–1080p, composite/S-Video/YPbPr/RGB/HDMI, genlock), discrete high-speed analog, power distribution, ngspice/SPICE and IBIS, DipTrace, Eagle, KiCad + own fork, DFM, BOM cost engineering, bring-up, oscilloscope/SCPI automation
Product & operations	Concept-to-production ownership, PLM (MRPeasy, InvenTree), ERP, supplier and contract-manufacturer management, pricing, documentation, open-source hardware programs
Mechanical	Inventor, FreeCAD, build123d, sheet-metal enclosure design, injection-molded knobs, front-panel graphics, Blender product rendering

EDUCATION

- B.A., Radio, Television & Film — University of North Texas, 2006